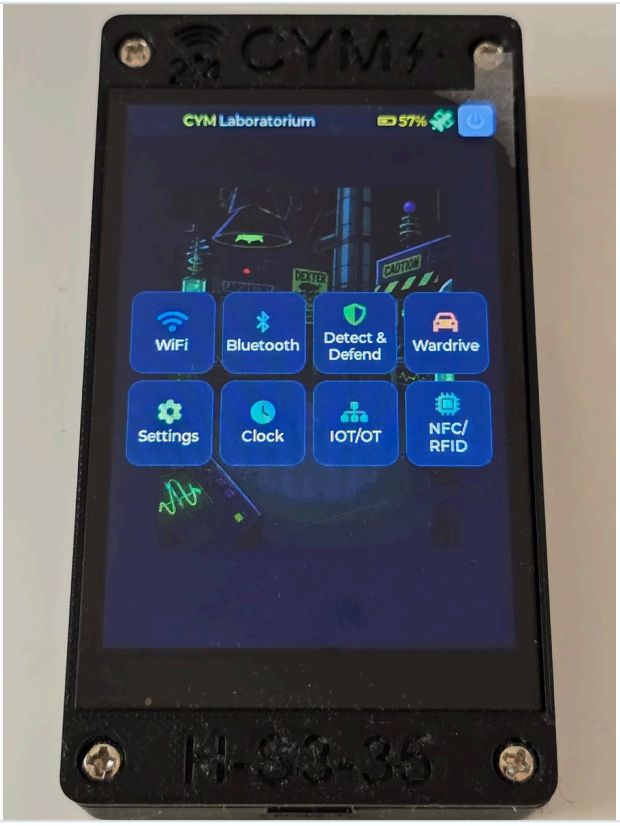
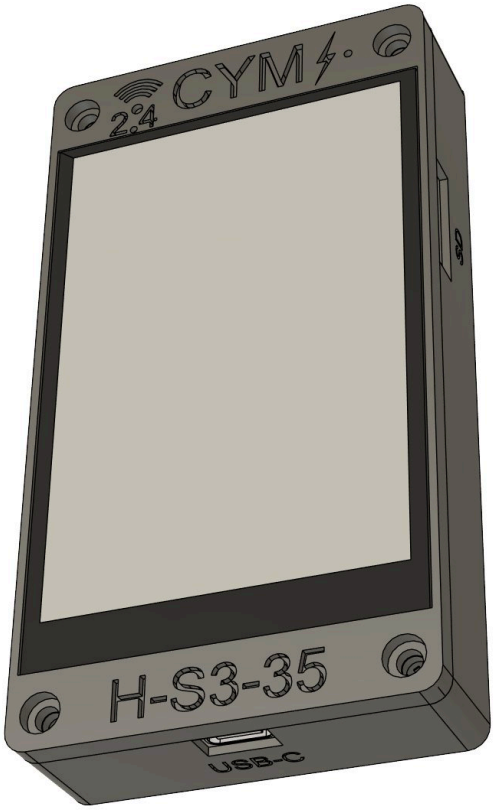
CYM Case Hosyond-S3-35 – 3D Printing and Installation Manual

Battery, GPS and speaker case for the Hosyond 3.5" ESP32-S3 touch display board (ES3C35P) · Standard and Slim · Revision 1

Designed and made by @birolt29 @alen29





The real device (CYM interface running) and the 3D design view (front).

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This manual has four parts: general information about the board and the parts used, 3D printing of the case and the front cover, assembling the parts into the case, and installing the firmware. Print settings are given by their generic names; you will find the equivalents in whichever slicer you use.

The case comes in two models; the front cover is the same for both. **Standard** (15.0 mm): with the speaker from the set, facing the grille in the bottom of the case. **Slim** (12.9 mm): no speaker, thinner. The battery, GPS module, GPS antenna and switch are the same in both. The photos show the Standard model; differences for the Slim are noted.



Finished device: front (screen on), socket wall (UART / I2C / EXP.INT), back of the printed case (Standard).



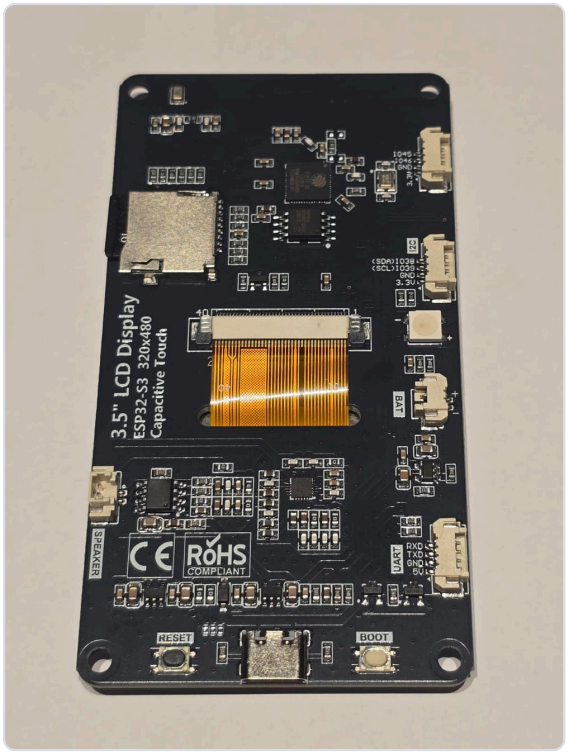
Design views: front (the same for both models), Standard back (with speaker grille), Slim back.

Part 1 – General information

Board specifications

Processor	ESP32-S3 · Xtensa LX7 dual-core · 240 MHz
Memory	16 MB Flash · 8 MB PSRAM
Wi-Fi	802.11 b/g/n · 2.4 GHz
Bluetooth	Bluetooth 5 (LE)
Display	3.5" IPS · 320×480 px · ST77922 driver (QSPI)
Touch	Capacitive · FT6336G (I2C)
Audio	ES8311 codec + FM8002E amplifier · speaker up to 1.5 W (8 Ω) / 2 W (4 Ω) · built-in MEMS microphone
RGB LED	WS2812B · IO40
microSD	SDIO interface
USB	USB-C (native ESP32-S3 USB: power + flashing)
Battery	3.7 V LiPo socket (BAT, 1.25 mm 2-pin) · on-board charger (TP4054, ≈290 mA)
Case dimensions	57.4 × 105.4 mm · Standard 15.0 mm, Slim 12.9 mm + front cover 3.9 mm = 18.9 / 16.8 mm in total

Connectors (pinout)



UART – GPS socket (4-pin, 1.25 mm)

Pin	Signal	Description
1	5V	5 V power output (GPS supply)
2	GND	Ground
3	TXD0 (IO43)	Board TX – GPS RX goes here
4	RXD0 (IO44)	Board RX – GPS TX goes here

BAT – battery socket (2-pin, 1.25 mm)

Pin	Signal	Description
1	BAT+	Battery positive (through the switch)
2	GND	Battery negative

SPK – speaker socket (2-pin)

Pin	Signal	Description
1	SP–	Speaker negative
2	SP+	Speaker positive

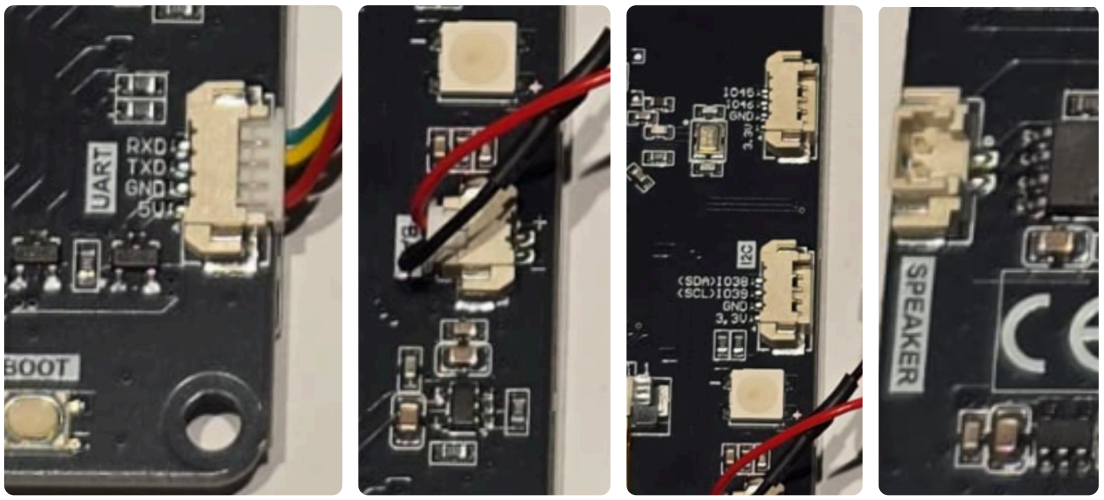
I2C – unused

Pin	Signal
1	3V3
2	GND
3	SCL (IO39)
4	SDA (IO38)

EXP.INT – unused

Pin	Signal
1	3V3
2	GND
3	IO46
4	IO45

Back of the board, USB-C at the bottom. The socket labels are printed on the board: UART RXD–TXD–GND–5V, EXP.INT IO45–IO46–GND–3.3V, I2C SDA(IO38)–SCL(IO39)–GND–3.3V, a + mark on the BAT socket, SPEAKER. Even so, check the battery polarity with a multimeter before plugging it in.



Socket labels on the board: UART, BAT (+), EXP.INT and I2C, SPEAKER.

Reference: parts and firmware used

Part	Product	Link
Board	Hosyond 3.5" ESP32-S3 IPS capacitive touch display (ES3C35P, 320×480); the set includes a speaker and a 4-wire 1.25 mm cable	www.amazon.de/dp/B0H2VPYP8N
Battery	3.7 V 1100 mAh LiPo with 1.25 mm plug. Sold as "YELUFT", arrives labeled "MakerFocus".	www.amazon.de/dp/B0FX8M8DK7
GPS module	ATGM336H GPS+BDS Dual Mode Module	www.amazon.de/dp/B0FKH26344
Switch	SS12D00G slide switch, 3 mm lever ("12D00G-3mm"); from the VEXUNGA 80-piece slide switch set	www.amazon.de/dp/B0FG2TSLFC
Screws + nuts	4 × M3 screws (6–7 mm under the head) and 4 × M3 brass hex standoff nuts (5 mm high); the same as used in the NM-CYD-C5 and 2432S028R cases, i.e., the ones supplied with those board sets. Not included in the Hosyond set; buy them separately.	–
Plug cable (optional)	Huba 1.25 mm 2-pin male + female plug cable pairs, 10 cm, 20 pairs	www.amazon.de/dp/B0GRGJBH13
Firmware	Cheap Yellow Monster (CYM)	github.com/JimGat/CYM#supported-hardware

The board has its own charging circuit, so no separate charge module is needed; the battery plugs straight into the board's BAT socket. The case has been tested with these parts. Before ordering, compare the battery size with the battery pocket: the pocket is 30.3 × 53.3 mm and 7.5 mm high, and batteries of the same capacity differ in size between manufacturers. Do not trust the listing either: the listing for the battery used here says 50 × 29 × 7 mm; the measured size is 53 × 30 × 7.5 mm. Links as of 2026-10-08; product pages may change. The firmware supports this board as "Hosyond ES3C35P 3.5"; select it as the board in the Web Flasher.

> Caution – battery polarity: with 1.25 mm battery plugs the polarity differs between manufacturers, and wire colors are not reliable either. Before connecting, check the positive terminal of the battery and of the board's BAT socket with a multimeter. Reversed polarity destroys the board's charging circuit.

Part 2 – 3D printing

Print files

File	Contents
`CYM_Case_Hosyond-S3-35_Standard.3mf`	Standard case + front cover together, in print orientation. Use this if your slicer can open 3MF.
`CYM_Case_Hosyond-S3-35_Slim.3mf`	Slim case + front cover together, in print orientation.
`CYM_Case_Hosyond-S3-35_Standard_Case.stl`	Standard case (with speaker). Units: millimeters.
`CYM_Case_Hosyond-S3-35_Slim_Case.stl`	Slim case (no speaker). Units: millimeters.
`CYM_Case_Hosyond-S3-35_Front_Cover.stl`	Front cover, the same for both models. Units: millimeters.

Two parts are printed for each model: the case and the front cover; no supports are needed. Standard case 57.4 × 105.4 × 15.0 mm, Slim case 57.4 × 105.4 × 12.9 mm, front cover 57.4 × 105.4 × 3.9 mm. Volume: Standard case 15.5 cm³, Slim case 14.0 cm³, front cover 5.2 cm³ (both parts together about 21–24 g of PLA).

Placement on the bed

- Case: the flat, closed back face (with BOOT / RESET / GPS) goes on the bed; the open side faces up.
- Front cover: the front face with CYM and H-S3-35 goes on the bed; the inner face (toward the board) faces up.
- The files are saved in this orientation; do not rotate them.
- Supports off. No supports are needed anywhere; the roof of the antenna pocket prints as a bridge.
- No brim is needed; add 3–5 mm if the corners lift.

Material

PLA or PLA+ is sufficient. Choose temperatures from your filament's label; because the part prints slowly, the lower-to-middle part of the range works well. Damp filament causes stringing and fills in small lettering; dry it before printing.

Slicer settings (0.4 mm nozzle)

Setting	Value	Why
Layer height	0.16 mm (first layer 0.20 mm)	For small lettering and holes. 0.20 mm also works, with a coarser result.
Walls (perimeters)	3	The side walls are 1.2 mm thick; 3 lines print them solid.
Wall generator	Arachne / variable line width	Thin walls and button tongues print in one piece.
Wall order	Outer wall first	Sharper outer surface and lettering edges.
Outer wall speed	30 mm/s	For the engraved lettering, the grille and the button slits.
Flow ratio	Calibrated; avoid over-extrusion	Over-extrusion narrows the button slits, lettering and nut pockets.
Top / bottom solid layers	4 / 4	
Infill	15%	
Elephant foot compensation	0.10 mm	Keeps first-layer squish from narrowing the lettering, grille and slits on the back face.
Supports	Off	

Setting names differ between programs (Cura, PrusaSlicer, OrcaSlicer, Bambu Studio, Creality Print and others). Reference print: an Ender-3 S1 class printer, textured PEI plate, 0.4 mm nozzle, PLA+, nozzle 210 / 205 °C, bed 60 / 55 °C; about 3 hours for both parts together (Standard), about 2 hours 30 minutes (Slim).

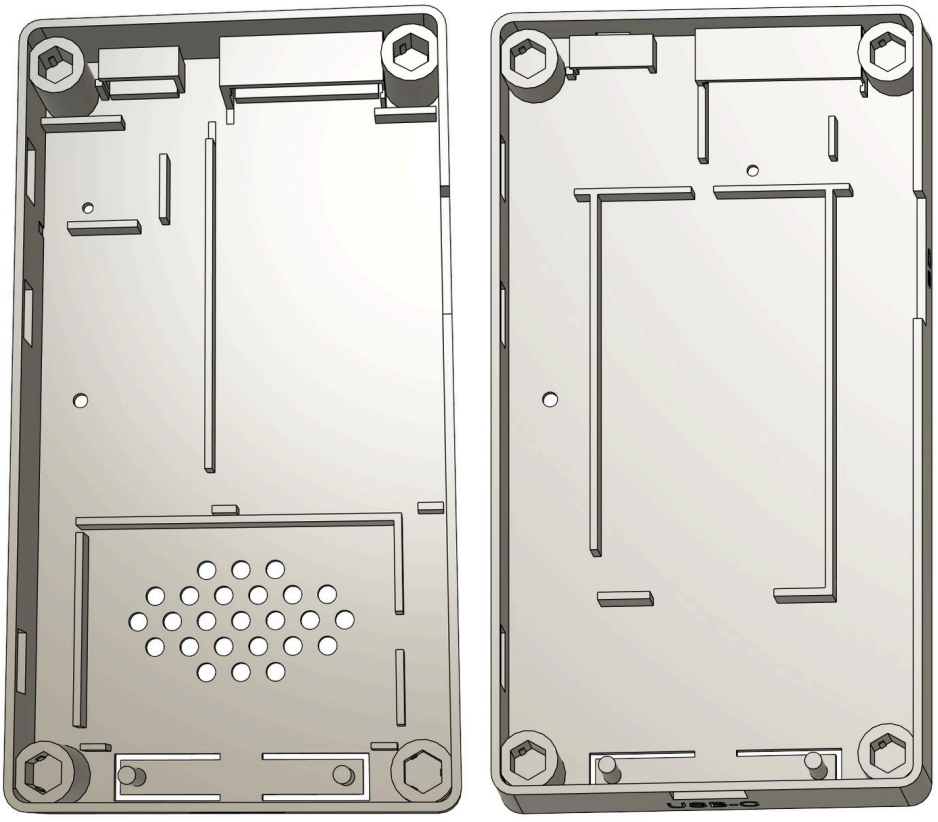
Checks after printing

- Do the BOOT and RESET tongues flex freely? If the slits along their edges are closed, open them with a craft knife.
- Is there sagging or a burr from bridging on the roof of the GPS antenna pocket? Clean it up before inserting the antenna.
- Are the hex nut pockets in the four posts clean, with the three small ribs in the lower half in place (they hold the nut)?
- Are the USB-C, UART, I2C, EXP.INT and SD openings and the switch slot clean?
- Are the 1.5 mm LED holes for the RGB LED and the GPS LED on the back face open? If not, open them by hand with a 1.5 mm drill bit.
- Standard: are the speaker grille holes open? Are there strings inside the engraved lettering?

Part 3 – Assembly

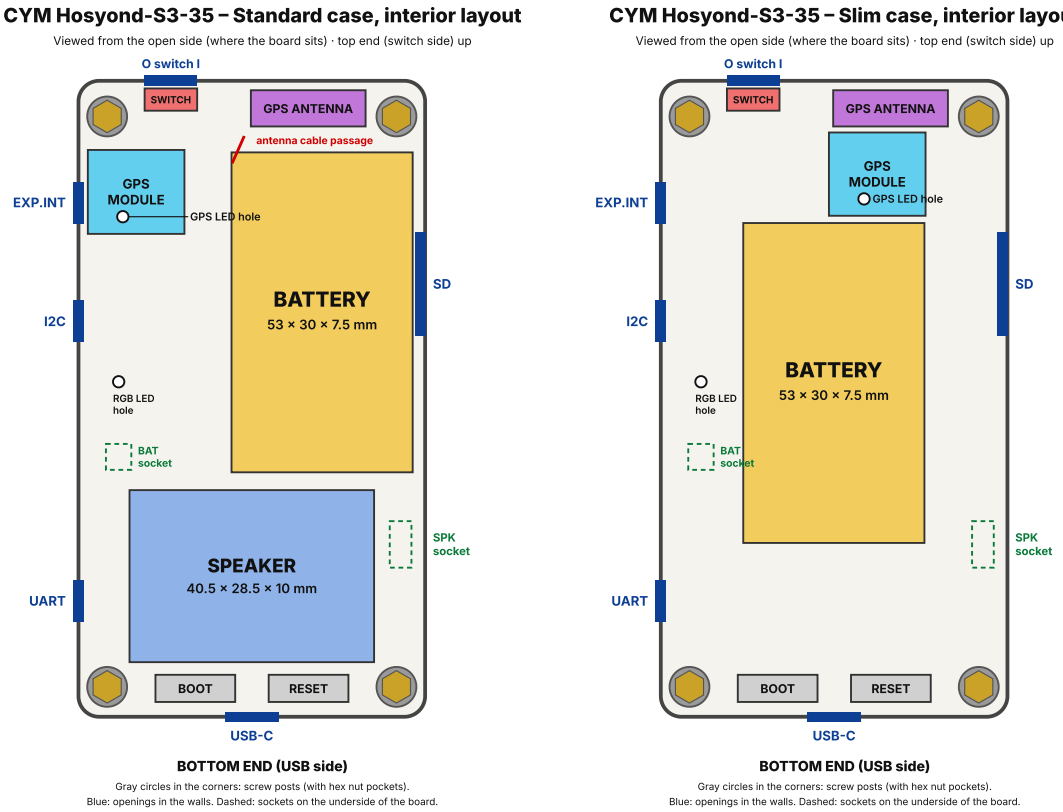
What you need

- Printed case (Standard or Slim) and front cover (CYM Case Hosyond-S3-35, Revision 1)
- Hosyond 3.5" ESP32-S3 board and what comes in its set: the speaker (40.5 × 28.5 × 10 mm, Standard only) and the 4-wire 1.25 mm cable (for the GPS)
- 4 × M3 screws (7 mm under the head, head diameter at most 5.6 mm) and 4 × brass hex nuts (5 mm high)
- LiPo battery 1100 mAh, at most 53 × 30 × 7.5 mm, 1.25 mm plug, protected
- GPS module 16 × 14 × 5.5 mm and its antenna (19 × 6 × 5.5 mm, 90 mm cable)
- SS12D00G slide switch, 3 mm lever
- Optional: 2-pin 1.25 mm plug pair (so the switch can be added to the battery lead removably)
- Thin double-sided tape, heat-shrink tubing, soldering kit, multimeter, tweezers, craft knife, hot glue



Inside the cases (design): Standard on the left, Slim on the right.

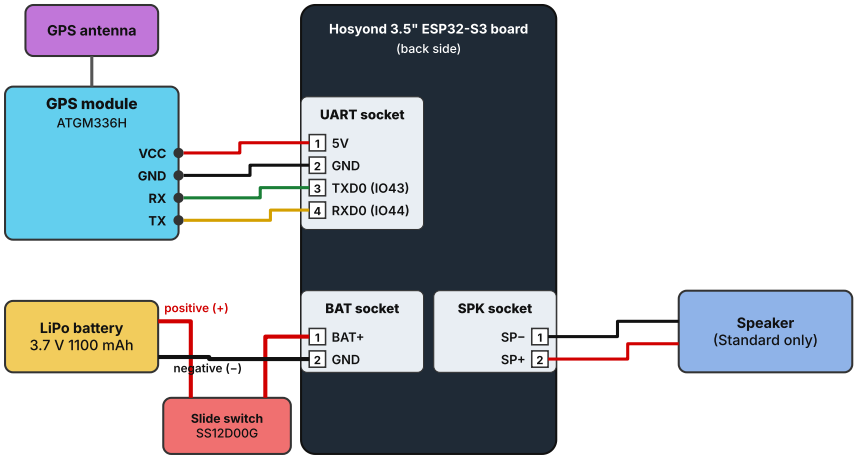
Interior layout



Component layout inside the case, Standard on the left, Slim on the right (viewed from the open side, top end up).

Wiring diagram

Which wire goes where: the battery goes through the switch to the board's BAT socket; the GPS module to the UART socket; the speaker (Standard) to the SPK socket. The antenna cable plugs into the small gold socket on the GPS module.



The switch is wired in series with the battery positive (+) lead. Go by the socket pin names, not by wire colors.

The battery is connected to the board only with the switch on (I); it must also be on for charging.

> Caution: wire colors of the GPS module and the battery differ between sets; although red + and black – are the usual convention, do not rely on it. Connect by pad/socket names, not by color (GPS VCC, GND, RX, TX ↔ UART 5V, GND, TXD0, RXD0; TX and RX cross over) and check the battery positive with a multimeter.

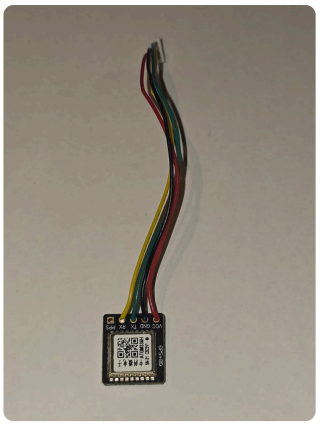
Preparation: soldering

Do these jobs at the workbench, before putting anything into the case.

Prep 1 Solder the cable to the GPS module

Keep the 1.25 mm plug end of the 4-wire cable from the set; that end goes into the board's UART socket. Cut off the 2.54 mm plugs at the other end and solder the four wires straight to the GPS module pads (VCC, GND, TX, RX); a pin header does not fit. Lead the wires off flat.

On the UART socket 1 = 5V, 2 = GND, 3 = TXD0, 4 = RXD0. GPS VCC goes to 5V, GND to GND, GPS TX to the board's RXD0, GPS RX to the board's TXD0. The ATGM336H runs on 3.3–5 V.



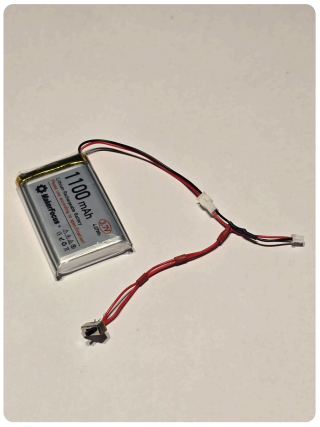


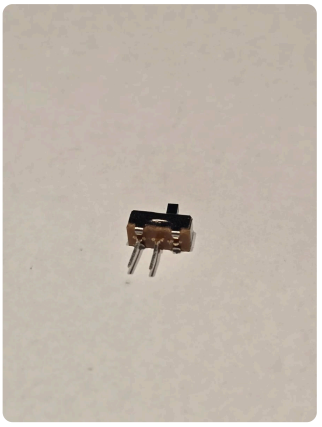
Prep 2 Add the switch to the battery lead

The switch goes in series with the battery positive lead: battery positive → middle pin of the switch, outer pin of the switch → positive wire to the board. The negative wire runs straight through. Cover the joints with heat-shrink tubing.

In the photo, a 2-pin 1.25 mm plug cable sits between the battery and the board, with the switch in its positive wire, so the battery lead stays uncut and the battery removable. Check the polarity of the board-side plug against the BAT socket (1 = BAT+, 2 = GND).

> Trim the switch pins; the pins and joints must fit into the switch pocket.





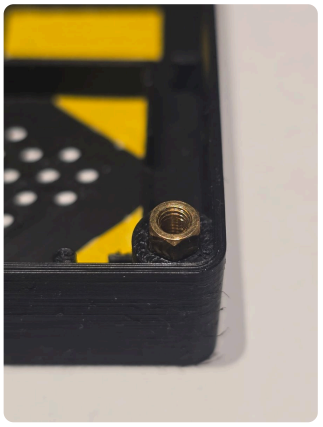
Assembly steps

Step 1 Fit the nuts

Press the four brass nuts into the hex pockets of the corner posts, all the way to the bottom, with the tip of a screwdriver. The small ribs in the lower half hold the nut; the nut must stay below the board and must not protrude above the post.

If a nut does not go down, clean the pocket; if needed, seat it by warming it with the soldering iron for 2–3 seconds.





Step 2 Apply double-sided tape

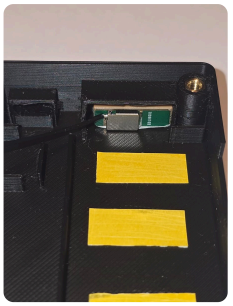
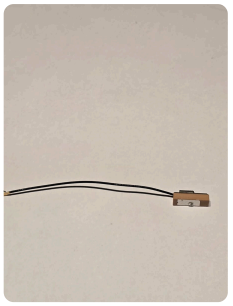
Stick three strips of thin double-sided tape on the floor of the battery pocket and one in the GPS module pocket. Standard: also one piece in each corner of the speaker frame; do not cover the grille holes.

> The tape in the GPS pocket must not cover the GPS LED hole in the floor.



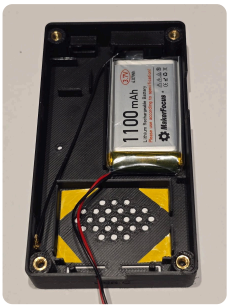

Step 3 Slide the GPS antenna into its pocket

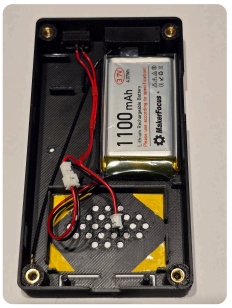
From inside the case, slide the antenna into the roofed pocket in front of the top wall; ceramic side down, cable exit to the left (switch side). The cable runs down through the passage at the left end of the pocket. Fix it with a drop of glue if needed.

Step 4 Fit the battery

Press the battery onto the tape in its pocket. Lay its lead, together with the switched plug cable (Prep 2), so that it reaches the board's BAT socket (on the left edge, just above UART); leave the plug free for now.

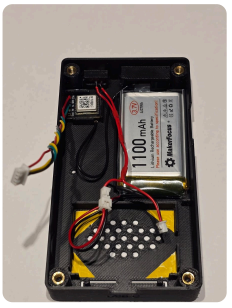




Step 5 Fit the GPS module and the switch

Press the antenna plug onto the gold socket on the GPS module. Press the module onto the tape in its pocket with the labeled (QR code) side facing you and the LED side toward the floor. Standard: top-left corner; Slim: top center, below the antenna.

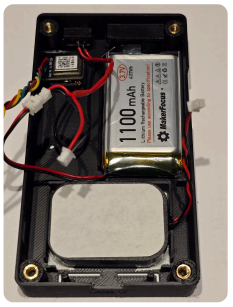
Seat the switch in its pocket at the top left; its lever comes out through the slit in the top wall, and position I is toward the antenna. The soldered wires leave the front of the pocket downward (red wires in the right-hand photo).





Step 6 Fit the speaker (Standard)

Press the speaker into its frame with the membrane facing the grille (the floor); the cable leaves through the gap in the middle of the right short side and goes to the board's SPK socket. The Slim has no speaker; skip this step.

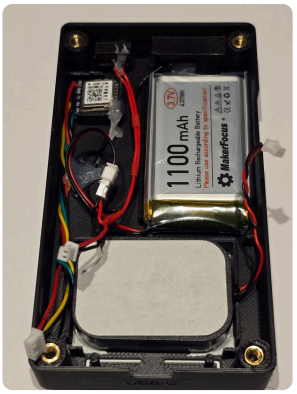


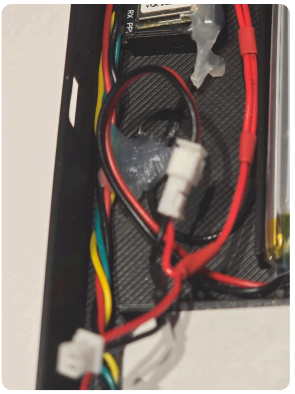


Step 7 Route the cables

Run the GPS cable along the left wall to the UART window. Lay the battery lead so it reaches the board's BAT socket (the inward-facing socket above UART). Standard: the speaker cable goes to the SPK socket on the right. Fold any extra cable beside the speaker frame.

Secure the wires with a few drops of hot glue. Keep glue off the BOOT / RESET tongues, their pins and the LED holes.

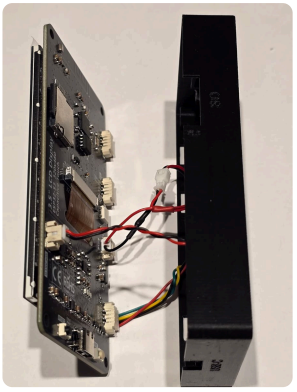




Step 8 Connect the board

Hold the board beside the case, screen down. Plug the GPS cable into UART, the battery lead into BAT and the speaker (Standard) into SPK.

> Test it before screwing on the front cover: turn the switch on (I). The board should power up and the screen backlight should come on. If the firmware is already installed, an image also appears; if not, see Part 4.





Step 9 Seat the board, screw on the front cover

Lower the board onto the four posts with the screen up; make sure no wire is trapped between the board and the posts. Place the front cover over the screen and fit the four M3 screws from the front. Do not overtighten.

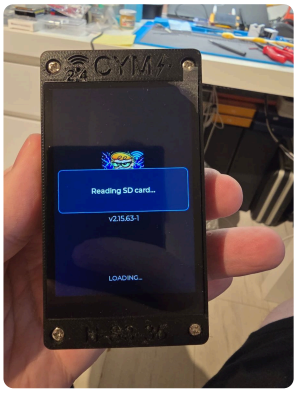


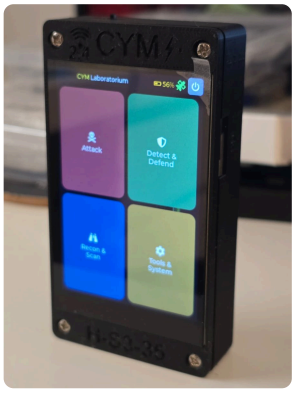


Step 10 SD card and final check

Insert the SD card through the SD slot in the right wall. The USB-C plug should go in easily; pressing the BOOT and RESET tongues on the back should click the buttons without leaving them pressed. The GPS LED shows through the hole in the back when the GPS is running.

> The battery charges through the board's USB-C socket (≈290 mA); the switch must be on (I) for charging.





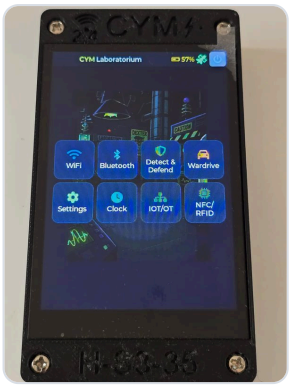
Part 4 – Firmware

Flashing the firmware with the Web Flasher

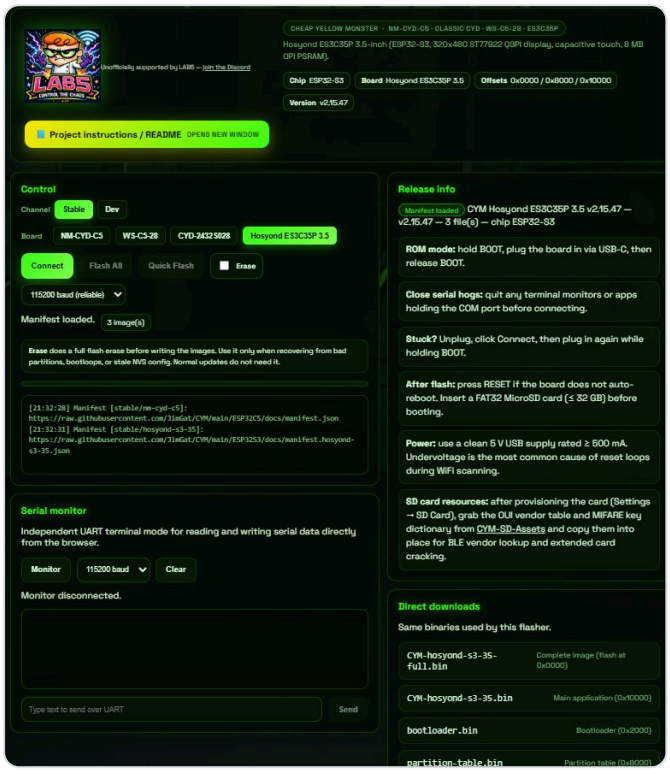
You can flash the firmware directly from the browser, without a USB driver or an IDE.

URL: https://jimgat.github.io/CYM/

1. Open jimgat.github.io/CYM/ in Chrome or Edge.
2. Select the "Hosyond ES3C35P 3.5" board.
3. Connect the board to the computer via its USB-C socket and click Connect.
4. Click "Flash" and wait until the upload has finished.
5. The board restarts by itself; the CYM interface appears on the screen.



When the upload is complete, the CYM main menu appears.



Web Flasher interface – "Hosyond ES3C35P 3.5" board selected.

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- **Not allowed without permission:** selling the printed case or the files, or using them commercially; publishing modified files.
- License text: creativecommons.org/licenses/by-nc-nd/4.0
- The CYM firmware is a separate project under its own license. Board and part names are trademarks of their manufacturers.

> The Web Flasher works only in Chrome and Edge (it needs the Web Serial API). Safari and Firefox are not supported.